

Long Run Market Access and the Rise of the Sunbelt*

Taylor Jaworski

Univ of Colorado, Boulder & NBER

Carl Kitchens

Florida State Univ & NBER

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Abstract

Abstract goes here...

*People to thank...

1. Introduction

As of 2019, over 82 percent of Americans live in urban areas, up from just under 40 percent in 1900. What explains the drastic movement of population over the 20th and 21st Centuries? For one, agricultural technology improved, changing the demand for inputs and altering patterns of crop specialization, freeing labor from the land Caselli and Coleman II (2001) Michaels et al. (2012) Olmstead and Rhode (2001) Griliches (1963) Jung (2020).

At the same time, cities have become healthier places to live and more productive. Early in US history, cities were dirty, unhealthy places that required a wage premium to attract workers to face risks associated with infectious diseases such as cholera, typhoid, yellow fever, and smallpox Cutler and Miller (2005) Beach et al. (2016) Ferrie and Troesken (2008) Alsan and Goldin (2019) Anderson et al. (2020).

As urban health improved, the need to attract workers with high wages subsided. While the amenity improvements theoretically dampen urban wage growth, cities have become incredibly productive places. As productivity has increased, the marginal revenue product of workers has increased, resulting in wage increases despite the improved city level amenities.

The urban wage premium has become an empirical fact, widely documented in the urban economics literature (see for example Rosenthal and Strange (2004) for a review). While documented repeatedly, the drivers of the urban wage premium are not fully understood. On the one hand, cities may attract workers that are more skilled Combes et al. (2008). For example Yankow (2006) documents that up to two-thirds of the urban wage premium is driven by differences in worker skills. Even after controlling for individual characteristics, the wage premium persists. Many authors have pointed to agglomeration economies to explain the remaining gap Glaeser and Maré

(2001) Wheaton and Lewis (2002).¹ Agglomeration economies are an attractive explanation, yet determining the underlying mechanism that contributes to agglomeration is difficult. Indeed a large literature has sought to explain the causes and mechanisms of agglomeration. In studying the urban wage premium, there is also a concern that the wage premium reflects firm survival, rather than agglomeration, although the existing evidence thus far points to agglomeration Combes et al. (2012);Behrens et al. (2014).

According to Marshall et al. (1920) physical proximity and connections to firms are a driver of agglomeration. Similarly, Duranton and Puga (2004) and Head and Mayer (2004) stress how trade costs may shape agglomeration and industrial location, and thus may partially explain the observed urban wage premium. In this paper, we document how changes in market access impact the estimated urban wage premium in the United States between 1940 and 2015. Empirically we begin by estimating regressions of individual level wages on a set of individual characteristics and location and time fixed effects. This set of regressions enables us to partial out the portion of the wage gap due to underlying differences in the skill-education-experience composition of workers between urban and rural areas, while at the same time, netting out common firm survival trends that contribute to the wage gap. Similar to prior work, we find that individual characteristics account for approximately half of the urban wage premium and become increasingly important after 1980. We then construct a measure of market access following Donaldson and Hornbeck (2016) and Jaworski and Kitchens (2019) at the MSA level drawing on newly digitized atlases of the US Highway Network each decade 1940-2010. Our measure of market access allows us to consider how changes in a cities location in the network affects the wage premium. We find that af-

¹Recently, work has also shown that urban industry-level concentration has led to a markdown of wages as firms exercise monopsony power Azar et al. (2020), thus, in the absence of concentration, the urban wage premium may be even larger than previously thought.

ter controlling for market access, the urban wage premium is cut in half. We interpret this finding as evidence that changes in trade costs that affect specialization patterns alter the demand for skills, explaining about a quarter of the overall urban wage premium. Our finding is consistent with hypothesis put forth by Duranton and Puga (2004) and Head and Mayer (2004) that firm industrial location decisions explain a portion of the wage premium.

While we consider the role that individual characteristics have on the urban wage premium, there is an underlying concern that individuals endogenously sort into specific cities based on unobservables, for instance, unobservable amenities or unobserved productivity differences. To address concerns regarding selection on unobservables, we follow Altonji and Mansfield (2018) and control for MSA level averages of individual characteristics in additional specifications. The inclusion of these MSA-level averages should, in theory, net out the influence of unobservable productivity, for example, due to higher returns to experience in urban areas, as described by Baum-Snow and Pavan (2012). For example, consider the unobservable trait of working longer hours, as discussed in Weinberg et al. (2004). If workers in a particular city work longer hours due to an unobserved and time varying preference for longer work days, then this may be reflected in the urban wage premium. Controlling for the MSA-level averages nets out this type of sorting, however it may overcontrol by limiting other the scope for other mechanisms. In this case, controlling for observables to address unobserved preferences for longer hours may limit the scope of referrals from neighbors, as discussed in Bayer et al. (2008). In addition to controlling for MSA-level averages of individual characteristics, we also estimate specifications where we control for MSA level housing values. We add housing value controls to limit the impact that unobserved amenities or land use restrictions have on wages.

When we add MSA-level averages of individual characteristics and MSA average housing values to our specifications, we find that the urban wage

premium rises/falls by XYZ. Overall, our findings highlight the role that selection, both by individuals and firms, contributes to the urban wage premium. Indeed, after controlling for many sources of selection, we find that the unexplained wage premium is XX percent of the observed premium.

In what follows, we describe how the urban wage premium has changed over time.

2. Historical Background

2.1 Highway Background

In 1940, there were 3.28 million miles of road in the United States.² Our data suggests that there were just over 358 thousand miles of federal and state aid highways in 1940. Of these, 126 thousand miles of US Highway were paved, 8.8 thousand miles were improved, but not paved, and over 2 thousand miles were dirt roads.³ The remaining 220 thousand miles of road in 1940 were state highways, of which just over 60 percent was paved.

Since 1940, federal and state highway networks have improved dramatically. First, over 1 million new miles of road have been added to the network, although much of this mileage is maintained by local governments and is for local travel.⁴ Of the 1 million new miles, approximately 50 thousand were built or upgraded to create the Interstate Highway System (IHS), which drastically increased travel speeds between places in the network. In addition to the construction of the IHS, other upgrades to state and non-IHS Federal-Aid Highways occurred, such as improved alignments, upgraded

²<https://www.fhwa.dot.gov/policyinformation/statistics/2018/vmt421c.cfm>

³Our measure of mileage may differ from other measures given our data procedure used to develop time harmonized highway networks. For example, rather than deleting segments and recoding segments over time, which generates 126,000 miles of paved US Highway, tracing 1940 paved US Highways results in 127 thousand miles.

⁴<https://www.fhwa.dot.gov/policyinformation/statistics/2018/vmt421c.cfm>

surfacing, and road widening.⁵ Today there are over 220,000 miles in the National Highway System, reflecting expansions and improvements of both the IHS and the US Highway network.⁶

Initial funds were allocated on the basis of mileage, land area, and population, but would later be allocated on the basis of a given state's cost share of the entire system.⁷ Therefore, states knew from the outset how much federal money would be allocated to build their segments of the network, regardless of when construction began during the bill's authorization. Since 1956, there have been numerous acts to add mileage to the system, improve mass transit investments, and continue funding the highway trust fund. Between 1956 and 1960, approximately 10,000 miles were integrated into the system from existing highways and new construction. Between 1960 and 1970, an additional 21,000 miles were constructed and opened to traffic, and between 1970 and 1980 about 10,000 miles were constructed. Today, approximately 50,000 miles of interstate highways crisscross the nation. Recent work highlights the aggregate value of these improvements to the highway network through reductions in trade costs, as well as the potential disamenities generated within cities Michaels (2008), Allen and Arkolakis (2014), Jaworski et al. (2020), and Brinkman and Lin (2019). For our purposes, the fact that states had to partially fund construction generates significant heterogeneity in the timing of construction and improvements to the network. Thus, even without accounting for endogenous population movements, there is meaningful variation in market access.

⁵Examples of non IHS improvements include the construction of the Appalachian Development Highway System, improvements to the Strategic Highway Network (STRAHNET), the Florida Turnpike, Texas Farm to Market (FM) Roads, among many others

⁶<https://www.fhwa.dot.gov/policyinformation/statistics/2018/hm14.cfm>

⁷<https://www.fhwa.dot.gov/publications/publicroads/96summer/p96su10.cfm>

3. Data

To estimate the relationship between the urban wage premium and the role that market access has on the premium throughout the post war era, we draw on several datasets. First, to measure market access, we construct a new harmonized decadal dataset of the domestic US Highway Network that we describe in more detail below. To measure the average earnings per worker (wages), we use the IPUMS Full Count Census for 1940, the 1 percent sample Census for 1950 and 1970, the 5 percent sample Census for 1960, 1980, 1990, and 2000, and the 2005, 2010, and 2015 American Consumer Surveys. Our sample consists of adult men living in the continental US between the ages of 25 and 65 who are employed for at least 26 weeks in the sample year. We apply IPUMS person weights in our summary statistics and all our regressions.

We compute average weekly earnings (wages) by finding total income divided by the number of weeks worked in the given survey year. For individuals earning the top income code, we set their total income to the lower top income cutoff multiplied by 1.5.⁸ If the urban wage premium is primarily driven by people in the top income group, our estimates of the urban wage premium will be biased downwards by this choice. All incomes are then multiplied by the Consumer Price Index to be in 2015 dollars. In order to have a consistent measure of occupation, we use the ‘occ1950’ variable provided by IPUMS which harmonizes occupation codes across all our survey years.

⁸These cutoffs are reported in IPUMS INCWAGE documentation. In 2015 \$, the cutoffs are \$84,500 for 1940, \$99,200 for 1950, \$199,500 for 1960, \$309,000 for 1970, and \$225,000 for 1980. Afterwards, IPUMS reports income as the median wage above that year’s cutoff.

3.1 Market Access Data

To construct a measure of market access, we follow the inter-regional trade literature developed in in Eaton and Kortum (2002), Redding and Venables (2004), and Donaldson and Hornbeck (2016).⁹ A county c 's market access, MA_c , is given by this system of non-linear equations

$$MA_c = \rho \sum_d \tau_{cd}^{-\theta} MA_d^{-1} Y_d, \quad (1)$$

where θ is the trade elasticity with respect to trade costs, ρ is a constant from the theoretical model, and Y_d is the income of county d measured by GDP.

Our measure of the iceberg trade costs to ship from county c to county d in year t , τ_{cd} , measures the fuel costs to travel to a county d as well as the wage paid to the truck driver for the drive time. The formula for τ_{cdt} is given by

$$\begin{aligned} \tau_{cdt} = & \text{distance in miles}_{cdt} \times \text{cost per mile}_t \\ & + \text{travel time in hours}_{cdt} \times \text{hourly wage of truck drivers}_t \end{aligned} \quad (2)$$

To solve the system of non-linear equations derived from the model requires detailed trade cost, τ_{cd} , from every origin to each destination. We approximate the origin-destination trade costs in each year by measuring travel costs through the contemporary highway network. Our measure of trade costs relies on measuring the distance and time traveled through the highway network, which is then monetized using hourly wages and fuel cost in each period, as highlighted above. To measure the time and distance post 2000 (2005 and 2015), we rely on the detailed shapefiles maintained by the federal government. For years prior to 2000 we georeference atlases produced by Rand McNally in each decade 1940-1990, as the federal govern-

⁹The particular trade model used is described in Jaworski and Kitchens (2019).

ment did not begin maintaining its database until 1994. To create a time consistent network, we begin with the 2000 version of the Federal Highway Administration’s shapefiles. Then, for each decade, we overlay scans of each Rand McNally atlas and edit the federal shapefile. Specifically, we delete road segments that were constructed between the current map year and 2000, and re-code segments that have been reclassified over time (i.e., segments of the US Highway network that became interstates, or paved roads that were previously gravel or dirt). We perform this procedure iteratively for each decade back to 1940. One advantage of this approach is that it locates roads based on modern GPS technologies to ensure accurate placement throughout time.¹⁰ We then follow the process outlined in Jaworski and Kitchens (2019) to measure the point to point distance travel time through the network each decade. That is, we represent each location by its geographic centroid, connected to the network by a set of access roads. We assign a travel speed to each segment in the network according to its classification (state, federal, interstate) and surface (dirt, gravel, paved), according to the parameters in 1. We then solve for the least cost route through the network for all pairwise combinations of origins and destinations.¹¹

The market access variable summarizes a county c ’s access to the transportation network by taking an average of other counties’ market size weighted by both the trade cost to ship to that market and by all other counties access to that market (MA_d^{-1}). Therefore a county c ’s market access is large when it has access to large markets that other counties have little access to.

To aggregate market access to Urban areas, we created a crosswalk of counties within MSAs. For each MSA, we average across the counties within

¹⁰One drawback is that this approach may ignore roads that were important prior to the modern stock of infrastructure.

¹¹More details on how the trade costs are calculated can be found in Section III of Jaworski and Kitchens (2019).

Table 1: Road Classifications and Speed Parameters

Class of Road	Speed limit	Speed limit
Interstate	65	70
US Highway (Paved)	45	55
US Highway (Gravel)	35	50
US Highway (Dirt)	30	45
State Highway (Paved)	25	45
State Highway (Gravel)	20	40
State Highway (Dirt)	15	35
Access Road	10	10

Notes:

the MSA.¹² Our measure of non-urban market access for a given state is the average of counties in that state that do not fall within any MSA. Therefore, for each individual we have a measure of whether or not they live in an urban area and the market access of the urban/non-urban area they live in. This process is detailed in Appendix B.

3.2 Summary Statistics

In Table 2, we report summary statistics of key variables. The first thing to note is that in the post war era, the share of people living in urban areas has steadily increased. Between 1940 and 2015, there has been a 15 percentage point increase in the urban share, from 66 percent in 1940 to 81 percent in 2015. This trend matches those found in Boustan et al. (2013). The

¹²Our results are robust to weighting by population. (NEED TO CHECK)

Table 2: Summary Statistics

Year	N	% Urban	log(Market Access)		Urban Weekly Wage (2015 \$)		Nonurban Weekly Wage (2015 \$)	
			Mean	SD	Mean	SD	Mean	SD
1940	16,939,585	0.66	8.20	1.59	33	20	25	17
1950	68,457	0.69	9.33	1.29	72	41	59	33
1960	1,364,970	0.66	9.92	1.24	124	77	100	60
1970	280,338	0.75	10.49	1.16	205	138	166	105
1980	1,910,440	0.73	10.70	1.04	409	284	341	221
1990	2,262,149	0.74	12.06	0.81	701	562	558	405
2000	2,583,088	0.79	12.35	0.74	1,008	1,015	767	677
2005	530,737	0.79	12.33	0.75	1,143	1,146	863	712
2010	530,783	0.80	12.29	0.75	1,228	1,210	961	819
2015	561,934	0.81	12.28	0.77	1,353	1,446	1,068	1,011

Notes: Summary statistics weights individuals by their person weight. Weekly wage is reported in 2015 \$. Market access is calculated from solving $MA_c = \rho \sum_d \tau_{cd}^{-\theta} MA_d^{-1} Y_d$, where transportation costs, τ_{cd} , are calculated as described by Equation 2.

log of market access also increases in the post war era, until about the year 2000, when it levels off. Market access increases in part due to major improvements in the transportation network, such as the expansion of the Interstate Highway System (IHS), Appalachian Development Highway System (ADHS), and other improvements in the state and federal aid highways that increase driving speeds. Aside from improvements in network infrastructure, the growth in market access reflects GDP growth, which has grown from 1.3trillion in 1940 to 18.2 trillion in 2015. The last four columns of Table 2 show the wage among people living in MSAs versus those that work in nonurban areas. Both urban and nonurban mean weekly wages are growing over time in 2015 \$ from \$33.24 and \$25.23 in 1940 to \$1353.95 and \$1067.48 in 2015.

Figure 1 shows within-year deviations of the log of market access from

the yearly mean at the MSA-level and the state average of non-urban counties. The figure shows that over time, non-urban areas have had a relative growth in market access, particularly in the southern and western United States. On the other hand, large cities in the 1940 had very large market access relative to the national average and this relative advantage declines over time from about 4 log points to about 2 log points above the mean. The deviation in log market access from the national average highlights the variation used in our regressions of the log of wage on the log of market access.

While the southern and western states have experienced relative gains in market access compared to the northeast, market access has increased everywhere throughout the 2nd half of the 20th Century. In Figure 2 we document how the log of market access has changed over time. For example, North Dakota in 2015 has about the same level of market access that New York City had in 1940.

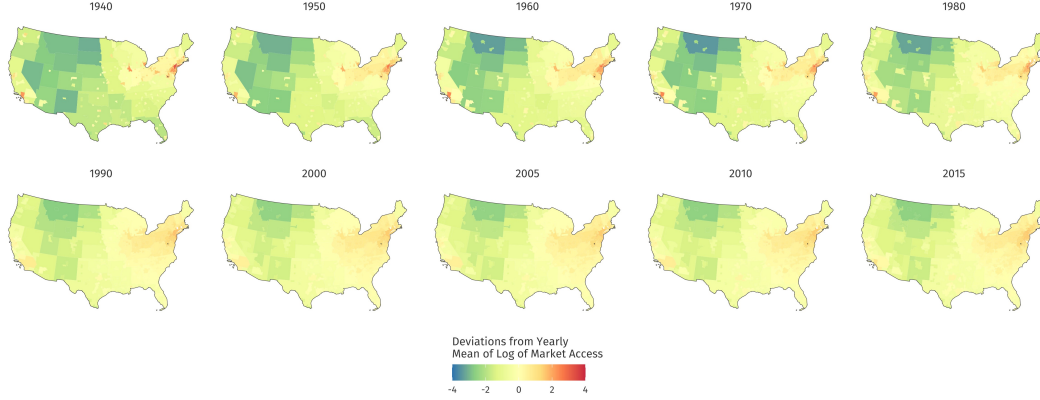
4. Results

For an initial understanding of the urban wage premium, for each survey year we take the ratio between the average urban weekly wage and the nonurban weekly wage. This ratio replicates the method used in Boustan et al. (2013) to calculate the urban wage premium. In Figure 3, the grey line with square markers shows the results of this exercise. The U-shape in which the urban wage premium starts around 30% in 1940, decreases between 1950 and 1980 to just over 20% and then increases again to over 25% after 2000 is consistent with results found in the literature.¹³

Moving into a regression framework, we run versions of the following

¹³Our results match the shape over time found in Figure 3 of Boustan et al. (2013). Our estimate in 2000 is a few percentage points lower than the estimate of 32% from Glaeser and Maré (2001). The increase between 1980 and 2000 matches Baum-Snow and Pavan (2012).

Figure 1: Log of Market Access Deviations from Yearly Mean: 1940-2015



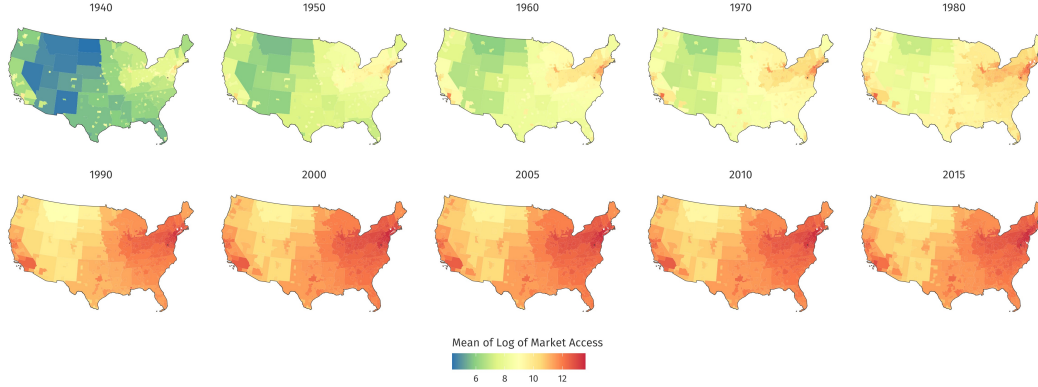
Notes: This figure shows deviations from the national mean of log market access averaged to MSAs and the state-wide average of non-urban counties. These averages are described in Appendix B. Market access is calculated from Equation 1 using trade costs calculated in Equation 2.

specification separately for each survey year:

$$\log(\text{Weekly Wage}_i) = \beta_0 + \beta_1 \text{Urban}_i + \beta_2 \log(\text{Market Access}_{MSA(i)}) + \delta X_i + \varepsilon_i, \quad (3)$$

where X_i is a set of controls consisting of age in 5-year bins, education levels, and an indicator for being White. Our estimate of β_1 is the percent increase in wages that workers in urban areas have on average which we call the urban wage premium. The first regression only includes an indicator for living in an urban area without the log of market access or controls X_i . This is the regression based version of the Boustan et al. (2013) approach. However, there can be demographic differences between urban and rural areas that contribute to wage differentials and so we add controls for education, race,

Figure 2: Log of Market Access over Time: 1940-2015



Notes: This figure shows the log market access averaged to MSAs and the state-wide average of non-urban counties. These averages are described in Appendix B. Market access is calculated from Equation 1 using trade costs calculated in Equation 2.

and age in the following specifications.

The wages of individuals in Urban areas also benefit from the improved transportation network, so in the final specification, we add in the log of market access. After controlling for market access, the coefficient on Urban, β_1 should pick up on the non-transportation causes of the urban wage premium.¹⁴

Our estimates for the urban wage premium will be biased if there is sorting into cities by characteristics that affect wages but are distinct from age, race, and education. An example could be that workers sort into cities based on the industry they work in and differences in industrial wages can explain some of the remaining wage premium. Combes et al. (2008) show using

¹⁴This includes the improved human capital development from Glaeser and Maré (2001) and improved firm-worker matching found in Baum-Snow and Pavan (2012).

panel data that individual fixed effects remove a sizeable portion of the variation in urban wage premium. Our time-series data is cross-sectional, so we can not use individual level fixed effects in our estimation. For our estimates to be causal, the exclusion restriction requires there to be no sorting beyond age, race, and education levels.

The results from the regressions that do not include a measure of market access is found in Figure 3 as the black lines with a diamond and a triangle marker. All regression tables can be found in Table A.1. These lines show that the urban wage premium still decreases from a premium of around 30% in 1940 through 1980. However, instead of the increase in the modern economy, our regression-based results show that the urban wage premium roughly levels off after 1980 at 20% without controls and at around 16% when controlling for age, education, and race. This result suggests that increased sorting of high-earning individuals into cities can explain the recent increase in urban wage premium found in the raw-measures.

The final specification includes market access in the regression of the log of weekly wage on an urban indicator variable. This specification controls for the benefits cities receive from being located in an advantageous location in the transportation network. If part of the wage premium is explained by better location in the transportation network, we expect the premium to decrease when including market access which summarizes the network. The estimates on the urban indicator are found in Figure 3 as the black line with circle markers and in Table 3.

When controlling for the transportation advantage of cities and demographic differences, we see the urban indicator in 1940 has decreased to about 15% from the 30% observed in specifications that don't include market access. There still exists, albeit a much smaller, decline in urban wage premium through the 80s which flattens out to an urban wage premium that floats around 10% until at least 2015.

This result suggests that between 25.9% to 47.7% of the urban wage premium is explained by a relatively strong position in the transportation network.¹⁵ In particular, it appears that early in our sample, the differences in access to transportation network was large between urban and nonurban areas. This matches the fact that historical highway networks focused on connecting large cities and starting around 1955 began connecting many more places which increased market access for less populous areas (as evidenced by Figure 1).

Table 3: Urban Wage Premium Estimates by Survey Year

	1940	1950	1960	1970	1980	1990	2000	2005	2010	2015
Urban	0.147 (0.010)	0.137 (0.012)	0.130 (0.011)	0.098 (0.011)	0.086 (0.011)	0.100 (0.011)	0.116 (0.011)	0.112 (0.011)	0.085 (0.014)	0.076 (0.017)
$\log(MA)$	0.053 (0.003)	0.035 (0.007)	0.051 (0.010)	0.056 (0.009)	0.038 (0.010)	0.093 (0.011)	0.070 (0.010)	0.068 (0.013)	0.067 (0.015)	0.054 (0.016)
N	16,939,585	68,457	1,364,970	280,338	1,910,440	2,262,149	2,583,088	530,737	530,783	561,934

Notes: Each cell is the coefficient estimate corresponding to the regression for the survey year indicated by the column label. The dependent variable is log weekly earning and each regression includes an indicator for living in an urban area, our measure of log market access, and controls for age in 5-year bins, education levels, and an indicator for being White. All regressions are weighted by the IPUMS person weights. Standard errors are in parenthesis and are clustered at the MSA-Nonurban State level

(@QUESTION: Not sure how to word the standard errors. They cluster at the MSA level and also at the nonurban state level.)

¹⁵The % change in the urban wage indicator between the specification including controls and the specification including controls and market access is 47.7% for 1940, 30.7% for 1950, 35.9% for 1960, 46.2% for 1970, 35.4% for 1980, 37.4% for 1990, 26.1% for 2000, 25.8% for 2005, 33.1% for 2010, and 27.4% for 2015.

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A. Full Urban Wage Premium Regressions

Full results of our regression estimates in Figure 2 can be found in Table A.1. In Panel A corresponding to the dotted line with diamonds in Figure 2, this set of regressions only have an indicator for living in an Urban Area. In Panel B corresponding to the dotted line with triangle markers in Figure 2, this regression adds controls for age in 5-year bins, education levels, and an indicator for being White. Lastly, Panel C replicates the full regression including controls and the log of market access which is the same as Table 3.

Table A.1: All Urban Wage Premium Estimates by Survey Year

	1940	1950	1960	1970	1980	1990	2000	2005	2010	2015
Panel A: Urban only										
Urban	0.3153 (0.0177)	0.2208 (0.0131)	0.2295 (0.0135)	0.2109 (0.0119)	0.1692 (0.0110)	0.1943 (0.0157)	0.2018 (0.0144)	0.2014 (0.0160)	0.1759 (0.0165)	0.1661 (0.0175)
N	16,939,585	68,457	1,364,970	280,338	1,910,440	2,262,149	2,583,088	530,737	530,783	561,934
Panel B: Urban with Controls										
Urban	0.3153 (0.0177)	0.2208 (0.0131)	0.2295 (0.0135)	0.2109 (0.0119)	0.1692 (0.0110)	0.1943 (0.0157)	0.2018 (0.0144)	0.2014 (0.0160)	0.1759 (0.0165)	0.1661 (0.0175)
N	16,939,585	68,457	1,364,970	280,338	1,910,440	2,262,149	2,583,088	530,737	530,783	561,934
Panel C: Urban, Controls, and Market Access										
Urban	0.147 (0.010)	0.137 (0.012)	0.130 (0.011)	0.098 (0.011)	0.086 (0.011)	0.100 (0.011)	0.116 (0.011)	0.112 (0.011)	0.085 (0.014)	0.076 (0.017)
$\log(MA)$	0.053 (0.003)	0.035 (0.007)	0.051 (0.010)	0.056 (0.009)	0.038 (0.010)	0.093 (0.011)	0.070 (0.010)	0.068 (0.013)	0.067 (0.015)	0.054 (0.016)
N	16,939,585	68,457	1,364,970	280,338	1,910,440	2,262,149	2,583,088	530,737	530,783	561,934

Notes: Each cell is the coefficient estimate corresponding to the regression for the survey year indicated by the column label. The dependent variable is log weekly earning and each regression includes an indicator for living in an urban area. Panel A just includes the Urban indicator; Panel B includes controls for age in 5-year bins, education levels, and an indicator for being White; Panel C includes the above controls and our measure of market access. All regressions are weighted by the IPUMS person weights. Standard errors are in parenthesis and are clustered at the MSA-Nonurban State level

B. MSA-level Market Access

We aggregate our county-level measure of Market Access from Equation 1 to a measure at the urban area. For each year, we start with a crosswalk provided by IPUMS that lists which counties fall (at least partially) inside the boundaries of the metro area. Our metro area boundaries are the 1950 Standard Metropolitan Area for 1940 and 1950, the Metropolitan Statistical Areas for their respective year in 1960-2010, and the Core-Based Statistical Area for 2015.

This list is modified slightly to include MSAs that are recorded in the IPUMS survey data but are not included in that year's crosswalk. The counties that the MSA covers are identified by hand using the shape files from NHGIS for that year or if missing from the previous year.

For a given year, a metro area's market access is the geometric average of the corresponding counties' market access. Then, for our measure of market access for the nonurban portion of a state is the geometric average of all counties that are not within any metro areas. We include robustness checks for weighting the average by county population.

B.1 Robustness Checks

B.1.1 MA Weighted by Population

For some metro areas, there is a central county and more counties at the peripheral, less populated counties. In this case, our simple average may understate the market access in a city, so we check for this by averaging across counties weighted by the counties population share. The results in Figure B.1 show that the shape of the black line with circle markers is very similar to Figure 3.

B.1.2 MA Without Own GDP

Since our measure of market access is increasing in a county's own size, there could be concern that our measure is not only measuring the county's access to the transportation network, but also the size of the county itself. ? finds evidence that the urban wage premium indeed is growing in city size. This would mean market access is absorbing some of the premium associated with being in a largely populated area and would be biasing our estimates of the wage premium towards zero. To avoid this, for all counties in an MSA, we construct a measure of market access that removes all counties from the summation that fall within the same MSA:

$$MA_c = \rho \sum_{d \notin MSA(c)} \tau_{cd}^{-\theta} MA_d^{-1} Y_d, \quad (4)$$

The results are in Figure B.2.

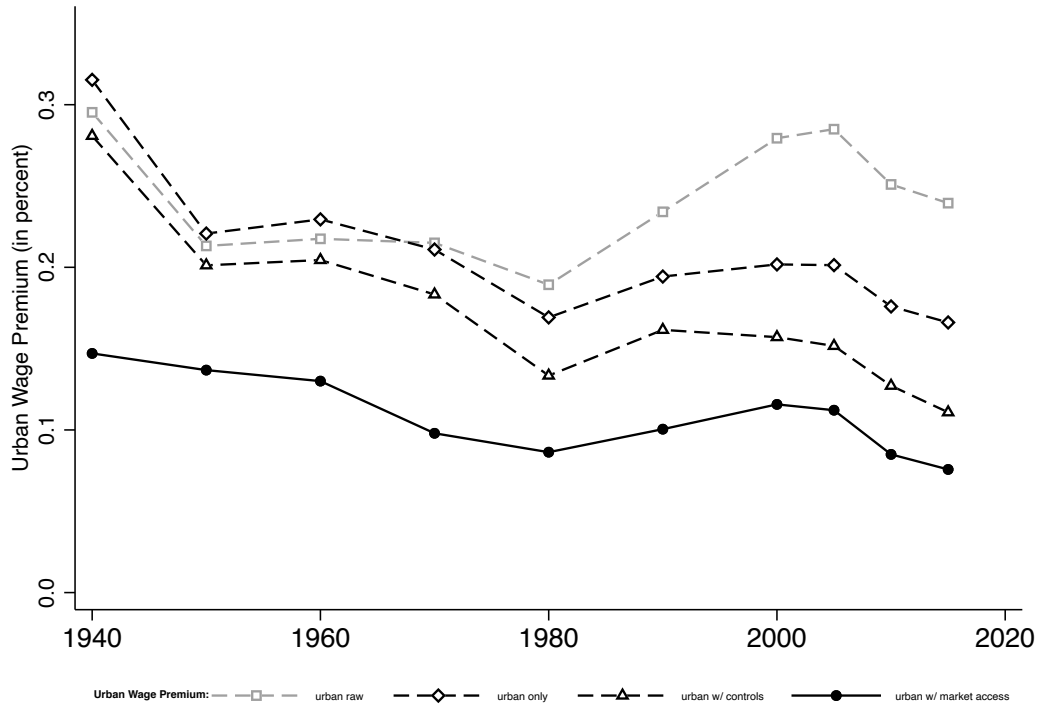
B.1.3 MSA Selection Over Time

Over decades as places grew in size, more and more places became classified as urban areas and these areas tended to be smaller than long-standing MSAs. There could be concern that the observed decline in the urban wage premium as smaller cities are added over time, the observed decline in the urban wage premium is driven by the addition of smaller cities over time. Figure B.3 shows this is not the case.

To create this, we keep all MSAs that exist in the entire sample from 1940 - 2015 (removing those that are added over time and the few that are lost over time). Observations are dropped for individuals identified in an MSA that does not exist for the entire sample. There still remains the issue that MSAs tend to grow over time incorporating less densely populated areas over time which we are unable to fix because the survey data does not feature exact

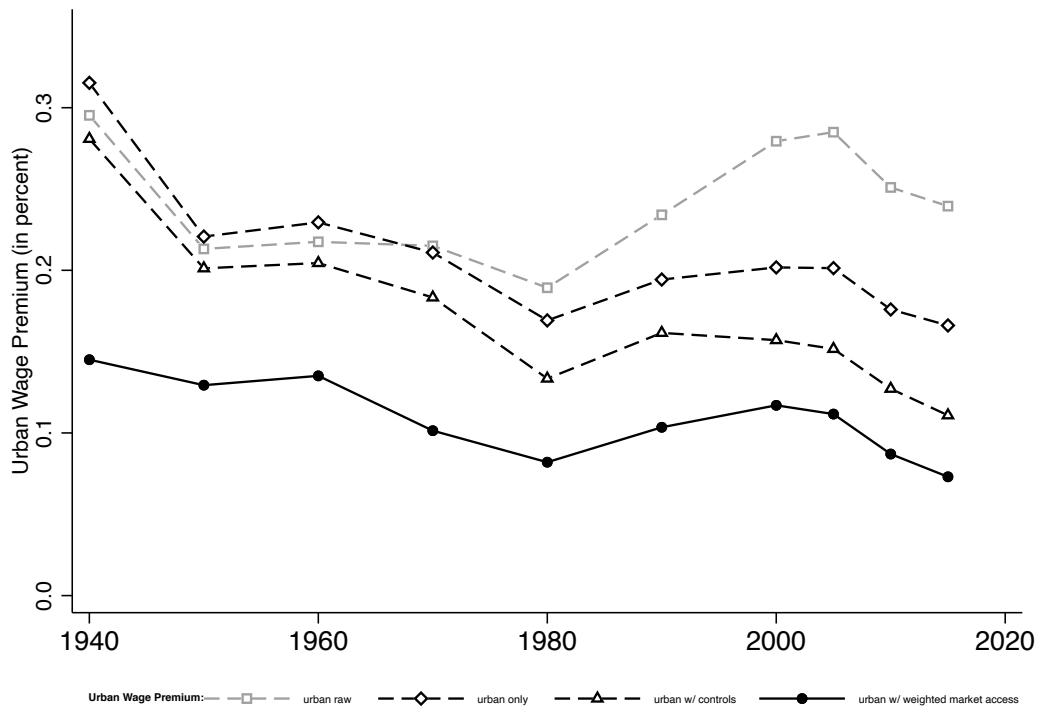
locations of individuals.

Figure 3: Urban Wage Premium 1940-2015



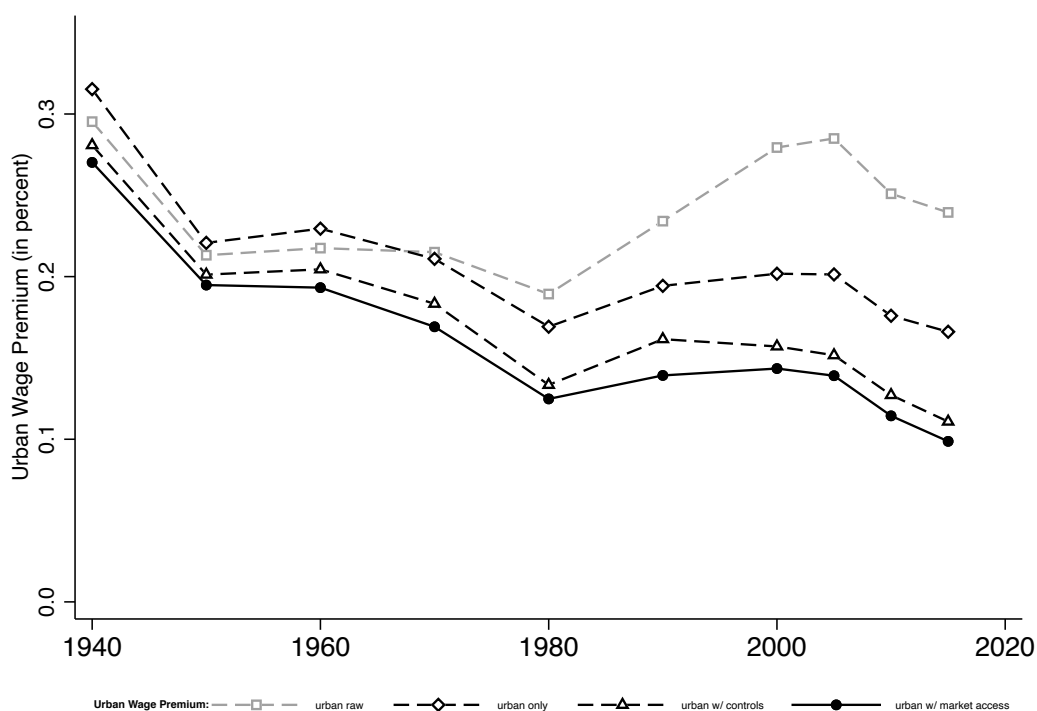
Notes: This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 3. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of market access.

Figure B.1: Urban Wage Premium 1940-2015 with Weighted Market Access



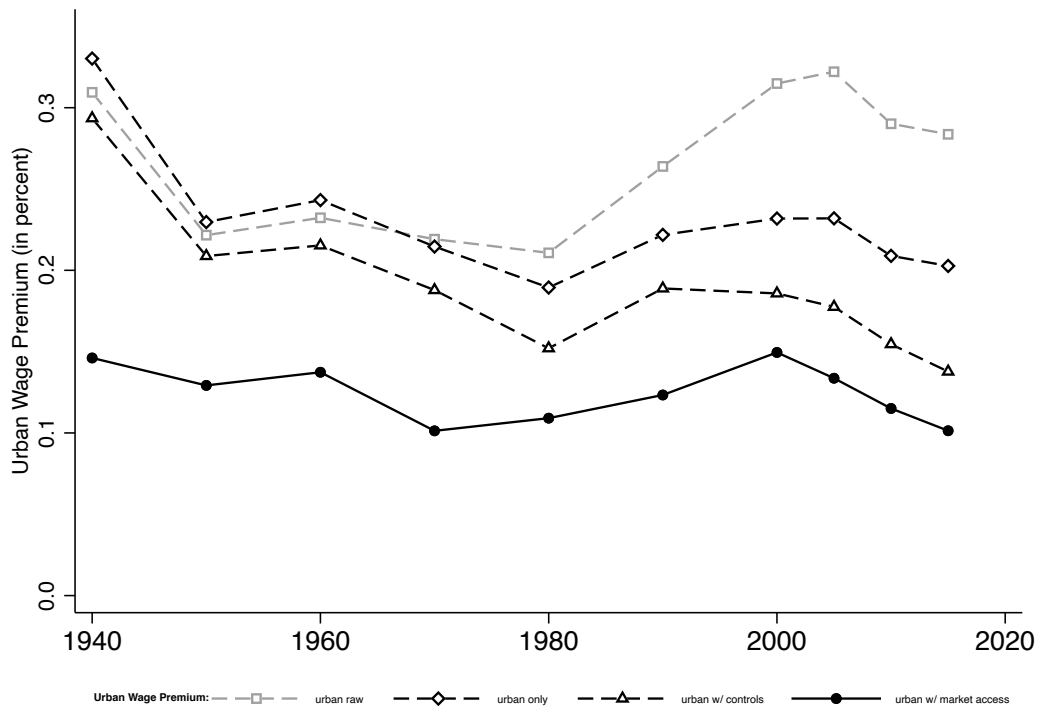
Notes: This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 3. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of the weighted market access.

Figure B.2: Urban Wage Premium 1940-2015 with Remove Own Market Access



Notes: This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 3. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of the weighted market access constructed according to Equation 4.

Figure B.3: Urban Wage Premium 1940-2015 with Weighted Market Access



Notes: This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 3. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of the weighted market access.